



Analytical Performance of a Fecal Extraction Device Versus Manual Extraction for Fecal Calprotectin Quantification in Inflammatory Bowel Disease

A. NASSAR (1), A. ALLIOUCH KERBOUA (1), H. MERDACI (1), S. GADIRI (1),
F. CHETTAB (2).

(1) Immunology Department, Sainte Thérèse Clinic, University Hospital Center (CHU) Annaba, Annaba, Algeria
(2) Gastroenterology–Hepatology Department, IBN SINA University Hospital Center (CHU), Annaba, Algeria

Introduction

Inflammatory bowel diseases (IBD), comprising ulcerative colitis (UC) and Crohn’s disease (CD), are chronic disorders characterized by recurrent episodes of intestinal inflammation. The current gold standard for evaluating intestinal inflammation is colonoscopy with mucosal biopsies. Although highly informative, this procedure is invasive, costly, time-consuming, and often poorly tolerated by patients.

Calprotectin is a calcium-binding protein located in the cytoplasm of neutrophils, accounting for approximately 60% of the proteins in the cytosolic fraction of these cells. In the presence of calcium, calprotectin is highly stable and remains intact in stool samples for up to one week at room temperature. Previous studies have demonstrated that fecal calprotectin is a reliable biomarker of intestinal inflammation and can serve as a useful non-invasive tool in the assessment and monitoring of IBD.

However, the analysis of fecal calprotectin can be labor-intensive, particularly due to a time-consuming extraction step. Following this extraction, fecal calprotectin is typically quantified using enzyme-linked immunosorbent assays (ELISA). The aim of the present study was to evaluate a novel fecal extraction method employing the Inova Diagnostics stool extraction device, in comparison with a conventional manual extraction procedure, under conditions in which both methods were performed in parallel on the same stool specimens.

Materials and methods

We conducted a prospective study including 23 patients with a previously established diagnosis of inflammatory bowel disease. The mean age of the study population was 40 years, and the female-to-male ratio was 1.3:1.

Each participant provided a stool sample, which was processed using two extraction methods: a manual extraction procedure based on stool weighing, and the Inova Diagnostics stool extraction device (Inova Diagnostics, a Werfen company), according to the manufacturer’s instructions. The resulting extracts were subsequently analyzed for fecal calprotectin concentration using an ELISA-based assay.

Calprotectin concentrations obtained from samples extracted with the device were compared with those derived from samples processed using the manual weighing method.

Statistical analyses were performed using Microsoft Office Excel 2016. Method comparison was assessed by calculating the correlation coefficient and evaluating the linear regression parameters (slope and intercept).

Results

Fecal calprotectin concentrations in the 23 samples processed using the manual extraction method ranged from 27 to 3000 mg/kg.

Comparison between the manual method and the novel extraction device yielded a regression slope of 0.67 (0.82–1.06) and an intercept of –4.5 (–4.9–0.6). The correlation coefficient between the two methods was 0.97 (Figure 1), indicating an excellent overall agreement.

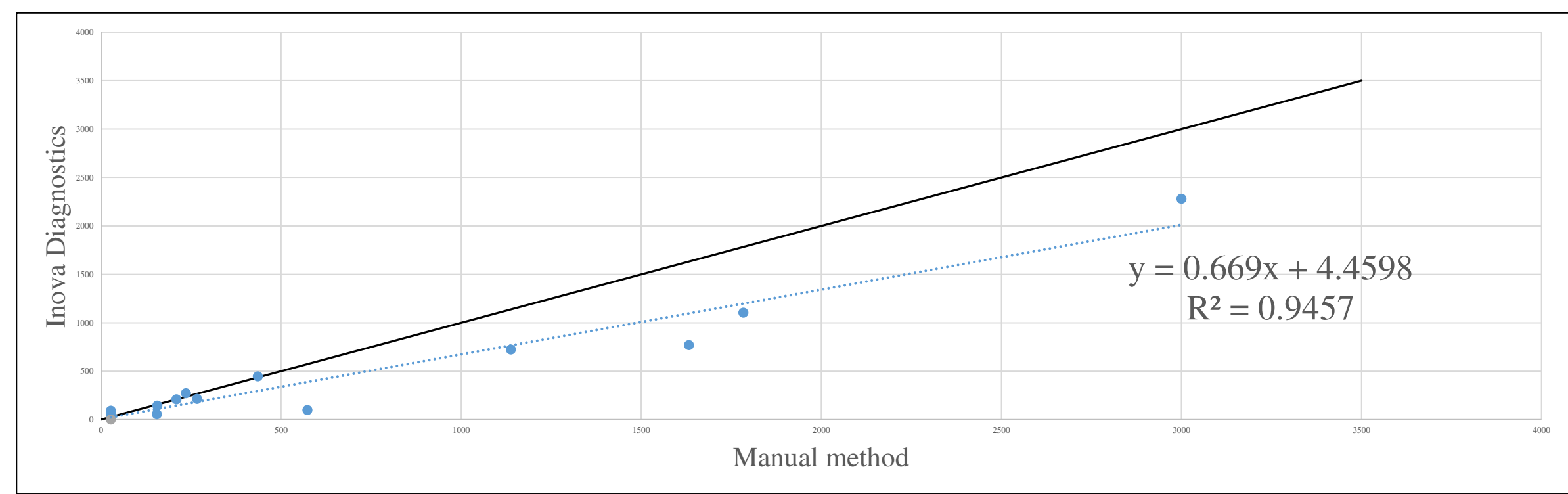


Figure 1. Comparative plot between the Inova Diagnostics stool extraction device and the manual extraction method for fecal calprotectin measurement. The solid black line represents the identity line ($y = x$).

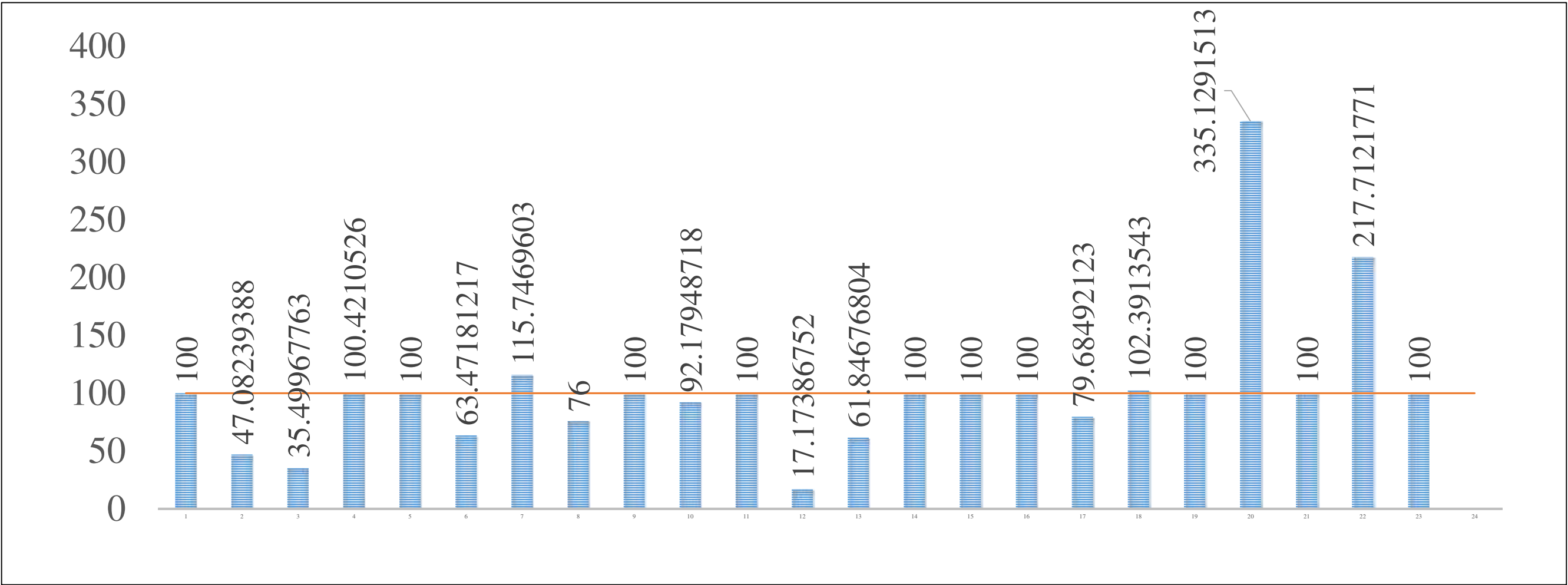


Figure 2. Comparison of the two extraction methods. The horizontal line represents perfect agreement with the manual extraction method.

Mean concentration (manual method)	Mean concentration (Inova Diagnostics device)	Mean bias
430,816	292,687	67,94 %

Discussion

In this study, we observed that the two fecal calprotectin extraction methods showed very good overall agreement, with a correlation coefficient of 0.945. This high degree of correlation indicates that, despite systematic differences, the commercial extraction device yields results that closely track those obtained with the manual reference method across the concentration range examined. However, extraction of fecal samples using the novel device resulted in globally lower calprotectin concentrations compared with the manual weighing method. The mean concentration obtained with the manual method was clearly higher, and the mean bias between the two procedures was 67.94%, corresponding to an under-recovery of 32.06% with the device. These findings are consistent with previous reports showing that the use of similar commercial stool extraction devices may lead to under-recovery of calprotectin of 28.1%, 25.4%, and 7.8% compared with manual weighing-based extraction. Sample 12 was particularly noteworthy, as it yielded a relatively very low concentration when processed with the device. This sample had a watery consistency, which likely affected the amount of stool actually collected by the device and, consequently, the measured calprotectin concentration. This observation underlines the potential impact of stool consistency on extraction efficiency and highlights the importance of considering pre-analytical factors when interpreting fecal calprotectin results obtained with such devices.

Conclusion

The use of the commercial stool extraction device resulted in under-recovery of fecal calprotectin compared with the manual reference method, which relies on accurate weighing of stool using a precision balance. Although the correlation between the two methods was excellent, the systematic negative bias observed suggests that results obtained with the device may not be directly interchangeable with those from the manual method. These findings should be taken into account when implementing commercial extraction devices in routine practice, and method-specific reference ranges or correction factors may be required to ensure appropriate clinical interpretation of fecal calprotectin measurements.

Références :

[1].Konikoff, M. R., & Denson, L. A. (2006). *Role of fecal calprotectin as a biomarker of intestinal inflammation in inflammatory bowel disease. Inflammatory Bowel Diseases*, 12(6), 524–534. doi:10.1097/00054725-200606000-00013

[2].Whitehead, S. J., French, J., Brookes, M. J., Ford, C., & Gama, R. (2012). *Between-assay variability of faecal calprotectin enzyme-linked immunosorbent assay kits. Annals of Clinical Biochemistry*, 50(1), 53–61. doi:10.1258/acb.2012.011272